Ezequiel Boero <ezequiel.boero@adagioinstitute.org>

Re: Subject: Selection Functionals Paper - My Response to Your Latest Comments

Benjamin D. Summers <bsummers@adagiollc.com> Tue, Sep 29, 2026 at 12:06 AM
To: Ezequiel Boero <ezequiel.boero@adagioinstitute.org>
Cc: "ISMAIL-SUTTON, SARA R." <sara.r.ismail-sutton@durham.ac.uk>

Thank you, Ezequiel. Your reading of Corollary 3.6 matches mine: (i) needs only $K_\Pi + \lambda K_\nu > 0$, which the unitriangular map gives for any real a, and (ii) uses $u > 0$ and the squares of the rates.

Your sweep violates the proof's inequalities; the proposition's conclusion holds in every case of it. Over the same family, $k_\nu \in \{0.5, 1, 2, 5\}$, k_Π from -1 to -20, $\lambda = 1/2$, $\sigma_\Pi = \sigma_\nu = 1$, $M = 200, 800, 3200$, the operator bound $C_M \le (\delta t/u)I$ fails in every $\kappa < 0$ case, by ratios 1.1 to 3.0, and $E_Q[V_M]$ converges to $\sigma_\infty^2 T = 2/3$ with $O(\delta t)$ error in every case, with $\mathrm{Var}_Q(V_M) = O(\delta t)$: at $k_\Pi = -20$, $k_\nu = 0.5$ ($\kappa = -19.75$) the mean is 0.6079, 0.6510, 0.6627 at the three grids, with error ratios 3.75 and 3.93 per fourfold refinement, and $M \cdot \mathrm{Var}$ tends to 0.88. So $V_M \to \sigma_\infty^2 T$ in Q-probability there too, and the conclusion of Proposition 7.5 holds for $\kappa < 0$. By Corollary 3.6(ii) the interior rows of the selected precision are those of an Euler chain with $a_{\mathrm{eff}} \in (0, 1]$ for either sign of the rates, so the selected chain is stable and its innovation variance tends to $\sigma_\infty^2 \delta t$ whatever the sign of κ. So $\kappa \ge 0$ is a hypothesis of this proof; the manuscript states it as a hypothesis of the proposition and scopes the necessity to $E \ge 0$, so nothing in it is wrong, and I leave it as it stands.

Ben

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